Sequence & Progression: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Arithmetic progression: basics

Formulas

$$T_n = a + (n-1)d, \qquad d = T_n - T_{n-1}$$

$$S_n = \frac{n}{2}\,[2a + (n-1)d] = \frac{n}{2}\,(a + l), \qquad n = \frac{l-a}{d} + 1$$

$$T_1 = S_1, \qquad T_n = S_n - S_{n-1}\ (n \geq 2)$$

KEY POINTS

- T_n linear in $n \Rightarrow$ AP; $S_n = An^2 + Bn$ (no constant term) $\Rightarrow$ AP with $d = 2A$
- Two terms given: $d = \dfrac{T_p - T_q}{p - q}$
- a, b, c in AP $\Leftrightarrow 2b = a + c$
- Assume 3 terms $a-d, a, a+d$; 4 terms $a-3d, a-d, a+d, a+3d$; 5 terms $a-2d, \ldots, a+2d$

Watch out

$S_n = An^2 + Bn + C$ with $C \neq 0$ is not an AP (only $T_2, T_3, \ldots$ are in AP).

2. AP: highlights and special problems

KEY POINTS

- Adding / multiplying every term by a constant keeps an AP; sum or difference of two APs is an AP; product is not
- Terms equidistant from the ends: $T_k + T_{n-k+1} = a + l$; $p + q = r + s \Rightarrow T_p + T_q = T_r + T_s$
- Ratio of sums $\dfrac{S_n}{S'_n} = f(n) \Rightarrow \dfrac{T_k}{T'_k} = f(2k-1)$
- Common terms of two APs form an AP with $d = \text{lcm}(d_1, d_2)$
- Multiples of m (or fixed remainder) form an AP with $d = m$; $\nwarrow a$ or $b\nwarrow$: inclusion–exclusion
- Greatest S_n: add terms up to the last positive term; smallest positive term: solve $T_n > 0$

Watch out

Count common terms only inside both ranges.

3. Arithmetic mean

Formulas

$$A = \frac{a_1 + \cdots + a_n}{n}, \qquad d = \frac{b-a}{n+1}, \qquad A_k = a + kd$$

$$A_1 + \cdots + A_n = n \cdot \frac{a+b}{2}$$

Watch out

n AMs between a and b give $n+2$ terms in all, so d has $n+1$ in the denominator.

4. Geometric progression and sums

Formulas

$$T_n = ar^{n-1}, \qquad S_n = \frac{a(r^n - 1)}{r - 1}\ (r \neq 1), \qquad S_\infty = \frac{a}{1-r}\ (|r| < 1)$$

$$0.\overline{pq} = \frac{pq}{99}, \qquad \text{sum of squares of an infinite GP} = \frac{a^2}{1 - r^2}$$

KEY POINTS

- No term of a GP is zero; a, b, c in GP $\Leftrightarrow b^2 = ac$

- Assume 3 terms $\frac{a}{r}, a, ar$; 4 terms $\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$; 5 terms $\frac{a}{r^2}, \ldots, ar^2$
- Product of terms equidistant from the ends is constant $(= a \cdot l)$
- Terms of a GP as triangle sides $(r > 1)$: $1 + r > r^2$

Watch out

The infinite sum needs $|r| < 1$; $r = 1$ needs $S_n = na$.

5. AP, GP and conversions

KEY POINTS

- a_n in AP $\Rightarrow c^{a_n}$ in GP $(c > 0)$; $b_n > 0$ in GP $\Rightarrow \log b_n$ in AP
- $a_{n+1} = pa_n + q \Rightarrow a_{n+1} + k = p(a_n + k),\ k = \dfrac{q}{p-1}$
- Terms of an AP that are in GP: use $T_q^2 = T_p T_r$ to get a in terms of d
- "Sum of squares $= 0$" $\Rightarrow$ each bracket $= 0$

6. Geometric mean

Formulas

$$G = (a_1 \cdots a_n)^{1/n}, \qquad r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}, \qquad G_k = ar^k$$

$$G_1 G_2 \cdots G_n = (ab)^{n/2}, \qquad G_k G_{n+1-k} = ab$$

7. AM-GM inequality

Formulas

$$\frac{a_1 + \cdots + a_n}{n} \geq (a_1 \cdots a_n)^{1/n} \ (a_i > 0), \qquad \text{equality} \Leftrightarrow a_1 = \cdots = a_n$$

STEPS

1. Make sure all variables are positive.
2. Minimum of a sum: choose parts whose product is a constant. Maximum of a product: choose parts whose sum is a constant.
3. Split with weights (e.g. $3x = x + x + x$, or $\frac{a}{5}$ taken 5 times) so that equality can occur.
4. Check that the equality point satisfies the given condition.

Watch out

If the parts cannot become equal, AM-GM gives only a bound, not the minimum.

8. Harmonic progression and harmonic mean

Formulas

$$T_n = \frac{1}{a + (n-1)d}, \qquad b = \frac{2ac}{a + c} \ (a, b, c \text{ in HP})$$

$$H = \frac{n}{\sum \frac{1}{a_i}}, \qquad \frac{1}{H_k} = \frac{1}{a} + kd, \quad d = \frac{\frac{1}{b} - \frac{1}{a}}{n+1}$$

$$\sum_{k=1}^{n} \frac{1}{H_k} = n \cdot \frac{\frac{1}{a} + \frac{1}{b}}{2}$$

KEY POINTS

- Every HP question: take reciprocals, solve the AP, take the reciprocal back

Watch out

The HP has no zero term; find where the AP changes sign to see when terms turn negative.

9. Relation between AM, GM, HM

Formulas

$$\text{RMS} \geq \text{AM} \geq \text{GM} \geq \text{HM}$$

$$G^2 = AH \quad \text{(two numbers)}, \qquad A = \frac{a+b}{2},\ G = \sqrt{ab},\ H = \frac{2ab}{a+b}$$

KEY POINTS

- For two numbers A, G, H are in GP; for $n > 2$ numbers $G^2 = AH$ holds only when the numbers are in GP
- Quadratic with roots a, b: $x^2 - 2Ax + G^2 = 0$

10. Arithmetico-geometric progression

Formulas

$$T_n = [a + (n-1)d]\, r^{n-1}, \qquad S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}\ (|r| < 1)$$

$$S_n = \frac{a}{1-r} + \frac{dr\,(1-r^{n-1})}{(1-r)^2} - \frac{[a + (n-1)d]\, r^n}{1-r}$$

STEPS

1. Write S and rS one below the other (shifted by one term).
2. Subtract: the middle becomes a GP; the AP part is reduced by one degree.
3. For an extended AGP (polynomial $\times$ GP) repeat until the AP part is constant.

11. Type 1: power sums

Formulas

$$\sum n = \frac{n(n+1)}{2}, \quad \sum n^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

$$\sum_{i<j} a_i a_j = \frac{(\sum a_i)^2 - \sum a_i^2}{2}$$

KEY POINTS

- T_n a polynomial in n: split and use the three formulas
- Alternating signs: pair the terms, or odd-placed and even-placed terms separately
- Patterns like $9, 99, 999$: $T_n = 10^n - 1$; $3, 33, 333$: $\frac{1}{3}(10^n - 1)$

12. Type 2: differences in AP / GP

STEPS

1. Check that $u_2 - u_1,\ u_3 - u_2, \ldots$ form an AP (or GP).
2. $T_n = u_1 + \sum(\text{differences})$; this gives T_n as a polynomial in n.
3. Add T_n using the type 1 formulas.

KEY POINTS

- Equivalent: for a quadratic $T_n = An^2 + Bn + C$ fit the constants from three terms

13. Type 3: method of difference

Formulas

$$T_r = f(r) - f(r-1) \ \Rightarrow\ S_n = f(n) - f(0)$$

$$\frac{1}{a_n \cdots a_{n+r-1}} = \frac{1}{(r-1)d}\left[\frac{1}{a_n \cdots a_{n+r-2}} - \frac{1}{a_{n+1} \cdots a_{n+r-1}}\right]$$

STEPS

1. Check: factors of the denominator in AP, numerator constant.
2. Write the numerator as (last factor $-$ first factor) $\div\, [(r-1)d]$ and split.

3. Write the first few and last few terms, cancel, keep the two end pieces.
4. Infinite sum: let $n \to \infty$.

Watch out

Use the step d of the factors (e.g. $d = 2$ for $r,\ r+2,\ r+4$).

14. Type 4 and continued products

Formulas

$$n(n+1)(n+2)(n+3) = \frac{1}{5}\left[n\cdots(n+4) - (n-1)\cdots(n+3)\right] \ \Rightarrow\ S_n = \frac{n(n+1)(n+2)(n+3)(n+4)}{5}$$

STEPS

1. Numerator is a product of r factors in AP: add the next factor, subtract the factor before the first, divide by $(r+1)d$.
2. Continued products: $T_r = c\,[P_{r-1} - P_r]$ with $P_r = \dfrac{1}{1\cdot 3\cdots(2r+1)}$ and so on.
3. Other splits: $\sqrt{k+1} - \sqrt{k}$, $\dfrac{1}{r!} - \dfrac{1}{(r+1)!}$, $\dfrac{r}{r^4+r^2+1} = \dfrac{1}{2}\left[\dfrac{1}{r^2-r+1} - \dfrac{1}{r^2+r+1}\right]$.

15. Recurrence relations and cubic roots

KEY POINTS

- $p = \sum a_n x^n$: multiply the recurrence by the right power of x, sum, solve for p
- Cubic with roots in AP: middle root $-\frac{b}{3a}$; in GP: middle root cubed $= -\frac{d}{a}$; in HP: put $x = \frac{1}{y}$
- Nested radicals $x = \sqrt{r+x}$: solve $x^2 - x - r = 0$ (positive root)

16. Groups, applications, sum of squares

KEY POINTS

- Groups: size of the groups $\to$ first term of the nth group $\to$ sum with S_n (sizes $2n-1$: first term $(n-1)^2+1$)
- Bouncing ball from height H with ratio e: distance $H + \dfrac{2He}{1-e}$
- Successive figures (midpoint triangles, squares on diagonals): perimeters, areas form GPs
- $(\dots)^2 + (\dots)^2 = 0$: both brackets are zero

17. Common mistakes checklist

STEPS

1. Number of terms in an AP: $n = \frac{l-a}{d} + 1$ (do not forget the $+1$).
2. S_n gives T_n only for $n \geq 2$; check T_1 separately.
3. n means between two numbers: denominator is $n+1$.
4. Infinite GP: check $|r| < 1$ before using $\frac{a}{1-r}$.
5. $b^2 = ac$ alone does not make a, b, c a GP if a term is zero.
6. AM-GM needs positive numbers and attainable equality.
7. HP: solve in the reciprocals, then invert back.
8. Telescoping: check the sign and the surviving end terms.
9. Common terms of two APs: keep only terms inside both ranges.